Lick Detection Board

(Alpha1d, Jan 2025)

Name	Value	Package	Description	Part Number
C1, C2	10 nF	0805	Input LP 5% C0G, NPO	490-1642-1-ND
C3, C4	.1 uF	0805	Trimmer filter	399-4264-ND
C5, C6, C9	.1 uF	0805	Comparator and buffer VCC Bypass	399-4299-ND
C7, C8	.1 uF	0805	Ref pin damping to improve stability (optional)	
DNP5_C, DNP6_C	.1 uF	0805	Output LP (Optional)	
~~D1~~	~~2A, 200V~~	~~Through hole~~	~~Input reverse polarity protection diode~~	~~RL203-TPCT-ND~~
D1	1A, 200V	SMA	Reverse polarity protection diode	S1D-FDICT-ND
D2, D3		Through hole SOD-123	Fast recovery Schottky diode for lick shock (optional, need to cut trace to utilize)	MBRS360BT3GOSCT-ND CMHD2003 TR PBFREE
Lick LED	Red	0805		**150080RS75000**
Power LED	Yellow	0805		**150080YS75000**
R1, R2	1k	0805	Input LP	
R3, R4	51k	0805	Threshold trimmer offset.	
R5, R6	4.7M	0805	Lick sense pull-up resistor. 4.7 is default value, but 1M or 10M can also be used depending on application	541-4.70MCCT-ND
R7, R8	2.4M, 1%	0805	Hysteresis divider	
R9, R10	1k	0805	Opto-isolator current limit	HMC0805JT50M0CT-ND
R11, R12	1.5k	0805	Ref pin damping to improve stability (optional)	RMCF0805FT2M40CT-ND
R13	1k	0805	PWR Led current limit	
RH_T, RH_B	100k 1%		Sets fixed hysteresis value if not using the trimmers	
PU1, PU2	10k	0805	Output Pull-up	
DNP1, DNP2	5M	0805	Pull down if using junction potential sense mode (optional)	
DNP3_R, DNP4_R	1k	0805	Output LP (Optional, would require cutting a trace)	
HYST_T, HYST_B	100k	Bourne 3362P	Hysteresis adjust (optional)	3362P-104LF-ND
TSH_L, TSH_R	100k	Bourne 3362P	Threshold pot	**3590S-1-104L** **3590S-2-104L**

		Bourne 3590 (10 turn)		
			Scale Dial for potentiometer	**H-516-6A** **2696**
OK1-OK4	Opto isolator	6-SMD	4N25SM	4N25SM-ND
IC2, IC3	Comparator	8-SOIC	LTC1540 ~~Max923 is 2 channel w/precision ref.~~	LTC1540CS8#TRPBFCT-ND ~~**MAX923CSA+**~~ ~~**LMP7300MA/NOPB**~~
IC1	TVS	SOT-143	TVS Protection diode for lick inputs. (Optional component)	CD143A-SR12CT-ND
IC4	Buffer non-inverting	SM8	SN74LVC2G126DCTR	296-12554-1-ND
	Right Angle BNC Double			ARF2111-ND ACX2286-ND
	Right Angle BNC			A97553-ND, WM5514-ND
	DC Barrel Connector			EJ503A-ND
	Micro USB connector		For optional power	609-4050-1-ND
	~~Switch 3PDT (On-Off-On)~~ ~~Switch 3PDT (On-On)~~		~~To select battery power.~~	~~360-3209-ND~~ ~~360-1864-ND~~
	~~Switch 4PDT~~		~~1004P3T1B1M1QEH~~	~~EG2429-ND~~
	Switch DPDT		500SDP1S1M2QEA	EG2475-ND
		~~SIP-7~~	~~Buckpuck 350 mA, internal current adjustment pot~~	~~788-1098-ND~~
			9V Battery holder	BA9VPC-ND

Version History:

Alpha 1a (Fabbed August 2023)

Based on the earlier MyRIO lick detection design.

1. LC at the input.
2. Can use reference voltage from comparator or 9V battery. Solder jumper to select.
3. Battery or wall power, selected by switch.
4. Provision for input diodes if delivering shock.
5. Can use conduction mode or junction potential mode.

Alpha1b (Nov, 2023)

1. Indicator LEDs on lick outputs?

2. mUSB footprint is not correct for the part I have
3. Bring out the hysteresis voltage set point on a test pad
4. Connect the 9V test point
5. Secondary footpring option for threshold pot

Alpha 1c (Fabbed Jan, 2025 PCBWay)

1. Line driver buffer on output
2. Lick indicator LEDs
3. Simpler power switch (Battery power for input stage is default, but can be changed by jumper)
4. Eliminated through hole diodes
5. Slight component placement changes

Alpha 1d (Fabbed Jan, 2025 PCBWay)

1. Fixed OE mistake on buffer
2. Updated jumpers and renamed few components
3. Assembly



